

Literature review and experimental methods

In this chapter a review of the field of van der Waals' materials is presented. Graphene's properties and prospects are discussed and the motivation for van der Waals' heterostructures outlined. Characterisation of graphene and heterostructures by different methods will be reviewed, with particular focus on techniques around transmission electron microscopy (TEM). FIB sample preparation for TEM imaging is reviewed, with particular attention paid to the challenges of this method when applied to beam sensitive 2D materials and its limitations. Common image processing and analysis techniques as applied to cross section images of heterostructures are discussed in the penultimate section. The final section gives an overview of density functional theory (DFT) calculations for van der Waals' heterostructures. DFT was carried out by coauthors of papers in the following chapters but was not part of the project. Nevertheless, it is the only technique which provides useful comparison to the quantitative STEM data presented in subsequent chapters, and so an understanding of it's merits and shortcomings is useful.

2.1 Graphene, van der Waals' materials and van der Waals' heterostructures

It is perhaps a hallmark of success that the name "graphene" means different things to different communities: from a strictly free-standing single atomic plane of carbon, to a vast field of research encompassing all atomically thin crystals utilising van der Waals' bonding. For the purposes of this work, graphene is a two dimensional allotrope of carbon: a single layer of carbon

atoms held together by sp^2 hybridised bonds to form a hexagonal lattice (see Figure 1).[60] These sheets can be sequentially stacked together to form bilayer, trilayer e.t.c graphene up to bulk graphite[30], wherein each graphene sheet is known as a basal plane.[6] Van der Waals’ materials can be defined as thick crystals composed of many layers of graphene or other 2D crystals, such as hexagonal boron nitride (hBN) and transition metal dichalcogenides (TMDCs). We will discuss 2D crystals later in this section.[61] It should be noted that bulk graphite, and all bulk crystals with homogenous composition and van der Waals’ bonding between atomically thin sheets, are included in the definition of van der Waals’ materials. An interesting extension of this definition occurs when different 2D crystals are stacked upon one another to form van der Waals’ heterostructures. We address these in the final part of this section. The papers presented in the following chapters characterise both van der Waals’ materials and van der Waals’ heterostructures, and not isolated sheets of 2D materials.

2.1.1 Graphene and van der Waals’ heterostructures

In 1947 P R Wallace attempted to explain the electronic properties of graphite by considering the electronic structure of the individual graphitic basal planes, finding each “to be a semi-conductor with zero activation energy”.[71] This work captured the essence of the unusual properties a single graphitic sheet would have, though they proved to be far stranger. In 2004 Nobel laureates Prof Andre Geim and Prof Konstantin Novoselov were the first to successfully isolate and characterise free standing graphene[60], although it should be acknowledged that monolayer graphite was known prior to this[32]. They

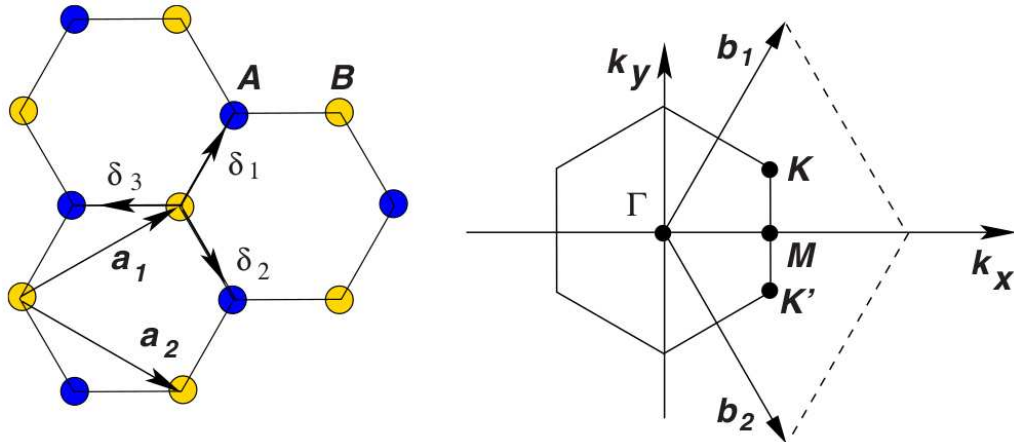


Figure 1: LEFT: Schematic showing the two sublattices (A and B) of graphene in blue and yellow, respectively. The lattice unit vectors, a_1 and a_2 are also denoted. RIGHT: The Brillouin zone of graphene. The Dirac cones are placed at the K and K' points. Figure reproduced from Castro Neto *et al.* Reprinted with (permission?) from American Physical Society, copyright 2009.

did this using the 'Scotch tape' method, where micromechanical cleavage is achieved using sticky tape to pull apart thick highly oriented pyrolytic graphite until it is atomically thin. The first electronic transport measurements not only showed that graphene's band structure has "Dirac cones" at the K-points of the Brillouin zone, and that charge carriers close to the centre of the cone behave as massless Dirac fermions, but that the anomalous quantum Hall effect and a non-trivial Berry's phase could be experimentally measured in high quality graphene flakes.[59, 80]

These phenomena all arise from two fundamental features of graphene's electronic structure. Firstly, charge carriers in graphene are chiral: Graphene has two sublattices which both contribute to the wavefunction of charge carriers, effectively burdening them with an additional quantum number "psuedospin". Psuedospin sets charge carriers in graphene apart from other

two dimensional electron gases as this additional property suppresses the backscattering of carriers. Broadly, this means that exotic quantum mechanical phenomena are more likely to be observed in graphene than in more traditional condensed matter systems where scattering can dominate measurements. A surprising manifestation of this effect is the Klein paradox, where carriers will always propagate through a sharp barrier if the angle of incidence is 0° to the barrier normal.[42, 79, 75] Graphene also has a unique electronic bandstructure: a gapless, linear energy spectrum for electrons and holes.[30, 18] Charge carriers in normal semiconductors can be described by a relatively straightforward Schrödinger equation, where the energy and effective mass varies parabolically. In graphene, the Schrödinger equation is not applicable to the linear energy spectrum. Instead the charge carriers must be described by a 2D relativistic Dirac equation as they are massless about the linear region; thus massless Dirac fermions. This is the unique case of monolayer graphene. In bilayer graphene an interesting combination of Dirac and Schrödinger equations results in massive Dirac fermions with four possible pseudospin states.

These revelations have propelled graphene to academic and mainstream fame, with the original Science paper receiving over 21,000 citations, placing it in the top 100 most cited papers.[70] Rapid progress in the field revealed monolayer graphene to have carrier mobilities of up to $200,000 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ at low temperature in the highest quality devices [20, 8], but, more interestingly for electronic devices, mobilities upwards of $15,000 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ at room temperature[32]. It was not long before it was realised that the thin silicon dioxide substrate, which was relied upon to locate graphene under the optical

microscope[7] and insulate it from the silicon back gate[59, 80], was breaking up the 2D electron gas in graphene through surface roughness, small areas of inhomogenous surface charging and impurities. The device performance was significantly improved by inserting a thin layer of insulating hexagonal boron nitride (hBN) between graphene and the substrate.[25] This heralded a step change in graphene device quality and was one of the first instances of a van der Waals' heterostructure.[31]

2.1.2 White graphene: the perfect dielectric

hBN is an insulating isomorph of graphene, sometimes dubbed “white graphene”, with strong in-plane ionic bonds in place of graphene’s covalent ones. Bulk hBN glues inert sheets of hexagonal boron and nitrogen lattices together with van der Waals’ bonds, similar to graphite. When thin sheets of graphene are placed on top of hBN and annealed, surface contaminants pool together in bubbles in what has become known as the self-cleaning phenomenon.[48, 38, 33] Self-cleaning arises from the van der Waals’ attraction between hBN and graphene[43] and results in large areas of atomically clean interfaces, proving vital for high quality measurements.[25] These humble properties have allowed some of the most exciting physical phenomena to be realised, including negative differential conductance[11], Hofstadter’s butterfly in bilayer graphene [24] and Klein tunneling of the chiral Dirac fermions.[42, 79, 75]

2.1.3 Potential applications of graphene

In addition to graphene’s incredible electronic merits are a host of other properties which are unrivalled by other materials. It is the thinnest imaginable

material (one atom thick)[60]; nearly completely optically transparent[56], has the largest specific surface area ($2630 \text{ m}^2 \text{ g}^{-1}$)[10]; the strongest and stiffest material ever measured (Young's modulus of 1 TPa and intrinsic strength of 130 GPa)[46]; has the highest thermal conductivity ($> 3,000 \text{ W mK}^{-1}$)[3]; the highest tolerance of electronic current density (a million times that of copper)[54, 55] and longest mean free path of electrons under ambient conditions[69]; the most stretchable crystal[37] and is impermeable to gases[15].

These myriad superior properties (most of which are only observed in the highest quality devices) make graphene an ideal candidate for many applications.[58, 29] Graphene-based inks and paints are expected to provide conductive inks, antistatic and interference shielding, corrosion barriers and gas membranes.[heap of refs] Transparent conductive coatings for consumer flexible electronics (e.g. flexible touch screens and rollable electronic paper) are already being developed commercially.[refs] By embedding graphene in polymer or ceramic matrices a new class of superstrong, lightweight and degradation resistant composites are anticipated to emerge over the next ten years.[rrreeefffsss] Graphene based composites are also poised to play important roles in boosting the performance of many electrochemical and optoelectronic applications such as energy harvesting and storage technologies (fuel cells, supercapacitors, batteries, etc.).[39] The properties of graphene are extremely sensitive to environment and can be used in many different sensor applications. These range from drilling holes in graphene in order to sequence DNA, to flakes acting as intrinsic strain gauges via Raman spectroscopy characterisation.[refs] Other sensing applications include gas environment, pressure and

magnetic field, with devices possibly measuring many of these metrics simultaneously.[reffs] Graphene is easily functionalised[ref] yet relatively inert which opens up possibilities in bioapplications, particularly in biomolecular sensing, tissue engineering, regenerative medicine and drug delivery.[ref]

It is widely anticipated that graphene will make it's biggest impact in the fields of electronics and photonics.[18, 2, 66, 5, 9] The poignant absence of a band gap in monolayer and bilayer graphene all but eliminates it's application as a speedy and direct replacement for silicon in conventional logic gate architectures. Nevertheless, there are currently three stratagies to open a band gap in graphene: confining the width of graphene to form nanoribbons[23, 47], biasing bilayer graphene[17, 62, 81], and straining graphene[63, 57]. Monolayer graphene nanoribbons currently show the most promise with bandgaps scaling inversely with the width of the ribbon, in some cases reaching 200 meV for < 20 nm widths[23, 47]. However, amongst problems relating to reproducible ribbon quality, confined carriers rapidly lose mobility and risk negating any advantages over silicon. The main motivation for integrating graphene in logic devices is to extend the trend in integrated circuit advancement as professed by the (largely self-fulfilling) Moore's Law, which in turn forms the basis of financial growth for the semiconductor industry.[53, 67] Through reduction in the size of silicon metal-oxide-semiconductor field-effect transistors (MOSFETs) the performance of integrated circuits has roughly doubled every 18 months for the last fifty years.[66] However size reduction is rapidly approaching it's limit in silicon and new materials are being sought to increase device performance and decrease feature size.[1] Although mobility is typically cited as graphene's most attractive feature for high speed logic de-

vices, it is actually the potential to have robust single atom channel widths which sustains interest within the semiconductor community (despite a lack of bandgap).[66] New device concepts are being explored to work around graphene's shortcomings, the most promising of which is the bilayer pseudospin FET; two graphene monolayers separated by a similarly thin dielectric. To switch the device on a gate bias is applied which encourages carriers to tunnel through the dielectric, effectively creating a short circuit current to flow through the channel. Under different bias conditions the tunnelling resistance is high, suppressing tunnelling and switching the device off.[4] This is anticipated to yield high on-off ratios with a very low switching energy.

The unique optical properties of graphene make it an excellent candidate material for many photonic devices. Photodetectors, mode-locked lasers, polarizers, isolator devices and optical modulators are all anticipated applications for graphene's optical properties, many of which already have proof of concept devices.[26, 40, 44, 21, 27, 36, 64, 74, 78, 52, 19] Optical modulators are key components in photonics systems as they are used to control the characteristics of light packets carrying signals between devices.[49] Modulation speed, the spectral range of the modulator, energy consumption and device size all dictate the quality of an optical modulator, and all of which depend on modulating material. Graphene is particularly useful for optical modulation for three reasons: Graphene has a high modulation speed owing to its intrinsically high carrier mobility; the absorption and modulation characteristics can be directly controlled by changing the Fermi level using a gate bias; and graphene has a large optical bandwidth with a constant absorption of 2.293% from the infrared to visible wavelengths (which happens to coincide

with the optical communication bandwidth required).[50, 64] This absorption may seem low, however when taken into consideration with graphene’s single atom thickness it is in fact very high compared to other photonics materials. These properties could prove most lucrative in terahertz modulators and detectors, which would begin to bridge what has become known as the “terahertz gap” and pave the way to cheap and powerful long wavelength applications.[40, 65, 52]

The main bottlenecks to progress in all these areas lie in the fundamental problems of handling graphene and scaleable synthesis.

2.1.4 Making and transferring graphene flakes

The papers in the following sections use devices made from mechanically exfoliated 2D flakes. For context a brief review of other synthesis routes to graphene flakes are presented here.

The myriad different methods for synthesising graphene flakes fall broadly under mechanical exfoliation, chemical growth and liquid phase exfoliation. Exfoliation methods are considered “top-down” synthesis routes as they directly remove graphene sheets from a bulk crystal of graphite. Chemical growth methods are “bottom-up” as they typically use a carbon-based precursor gas and favourable growth conditions (a catalyst, heat, pressure-control etc) to form sheets of graphene on a surface. The relative merits of each of these methods are best judged in the context of their target application. For example, liquid exfoliation methods are best suited to inexpensive large-volume production of graphene suspensions to generate conductive surfaces, composites and barrier films.[refs] However, liquid exfoliation struggles to make

atomically thin, clean, highly crystalline, large area flakes required for electronic and photonic devices (this is not for lack of trying[77]). Nevertheless, liquid exfoliation methods are easily the cheapest and most scaleable syntheses for graphene (and other 2D materials), and as such the applications which use liquid exfoliated graphene will be the first to be commercialised.[58, 29]

The highest quality devices, used for prototyping, proof-of-concept and exploring new physical phenomena, are still made from mechanically exfoliated flakes. Mechanical exfoliation is highly inefficient in terms of materials and time. Only a few suitable flakes are produced from a bulk crystal of graphite and, whilst the exfoliation step is not very time consuming, manually searching for flakes in an optical microscope is. Furthermore, the size and crystal directions of graphene flakes on a surface are not controllable by this method. Chemical growth methods

Almost regardless of how graphene is made, poor handling rapidly degrades flake quality as crystal defects, wrinkles and impurities set in. Handling strategies are largely the realm of surface science, with transfer techniques to and from different surfaces developed to aid the rapid but controlled transfer of graphene sheets from synthesis to target surfaces.

2.1.5 van der Waals heterostructure fabrication

such as these have largely only been possible due to progress in 2D crystal transfer techniques.[41] Broadly, there are two methods used to transfer one mechanically exfoliated 2D crystal on top of another: dry transfer and wet transfer, although other techniques are possible for different synthesis routes.[22] Wet transfer uses a sacrificial water soluble polymer layer to

transfer 2D flakes. The sacrificial layer is spin coated onto a silicon substrate and then further coated with polymethyl-meth-acrylate (PMMA). Graphite is mechanically exfoliated onto this polymer stack and thin flakes identified by optical microscopy[7] and Raman spectroscopy[28]. To isolate a suitable flake, the stack is floated on the surface of a water bath until the water soluble polymer dissolved, allowing the silicon substrate to sink. A glass slide with a cut hole in the middle is then manoeuvred with a micromanipulator over the flake on the floating PMMA to remove them from the bath, all observed under an optical microscope. The PMMA is flipped over so that the flake-side is facing the bench. The flake is aligned above a target substrate flake (e.g. hBN) and the two brought into contact to make a “stack”. The PMMA is then dissolved with acetone and washed away. The stack is annealed to $> 110\text{ }^{\circ}\text{C}$ to drive off solvents and contamination.

Dry transfer is a variation on the wet transfer technique which attempts to minimise contamination between flakes and greatly speed up the transfer process.[76, 72, 45] The wet transfer procedure is followed to the point where a flake is isolated on a PMMA membrane above a smaller target flake. Once the two flakes are in contact, instead of dissolving the PMMA, the PMMA is lifted from the substrate bringing the two flakes with it. This step is then repeated for many flakes to build up a bespoke complex stack of different 2D materials. On the last transfer step, the PMMA is dissolved with acetone and the stack annealed. Through this method the majority of interfaces between flakes will be isolated from PMMA and solvents.

Stacks of 2D crystals are made into devices by adding metallic contacts to the surface[11, 35, 34] or edges of graphene sheets[72, 45]. More recently, in the

latest generation of devices, metal contacts are placed on the surface of an encapsulating hBN layer so that electrons can be injected by tunnelling into buried conductive layers or used as a gate to probe the electronic properties of a device.[14, 12, 13, 33, 77, 16]

2.1.6 Transition metal dichalcogenides

The transition metal dichalcogenides (TMDCs) are metallic or semiconducting crystals of formula MX_2 , where M is a transition metal (e.g. molybdenum, tungsten, niobium, etc.) and X is a chalcogen (sulfur, selenium or tellerium). Similarly to graphene and hBN, a TMDC has hexagonal lattice sheets with strong in-plane bonding. Each sheet is stacked on top of another and held by van der Waals' forces. However, unlike graphene and hBN, they are not one atomic layer but three, with a sheet of transition metal atoms encapsulated by two sheets of chalcogen atoms. The most popular TMDCs are currently MoS_2 , WS_2 , MoSe_2 and WSe_2 due to them having an indirect semiconducting band gap in bulk crystals and a direct gap in monolayer crystals.[51, 68] The direct band gaps of the popular TMDCs range from $\sim 1.5 - 2.1$ eV[73], which correspond to near infrared - yellow photon energies in the visible spectrum. Furthermore, these TMDC monolayers are “multi-valley semiconductors”, meaning they have favourable and distinct localised states which a charge carrier wavefunction can occupy. These states appear as valleys at the K points of the Brillouin zone (K and K- in Figure 2), with differences between valleys arising from a lack of inversion symmetry in the crystal structure of monolayer TMDCs. A carrier can be confined to one of these valleys and so gains an extra constraint: a k -valley index (also

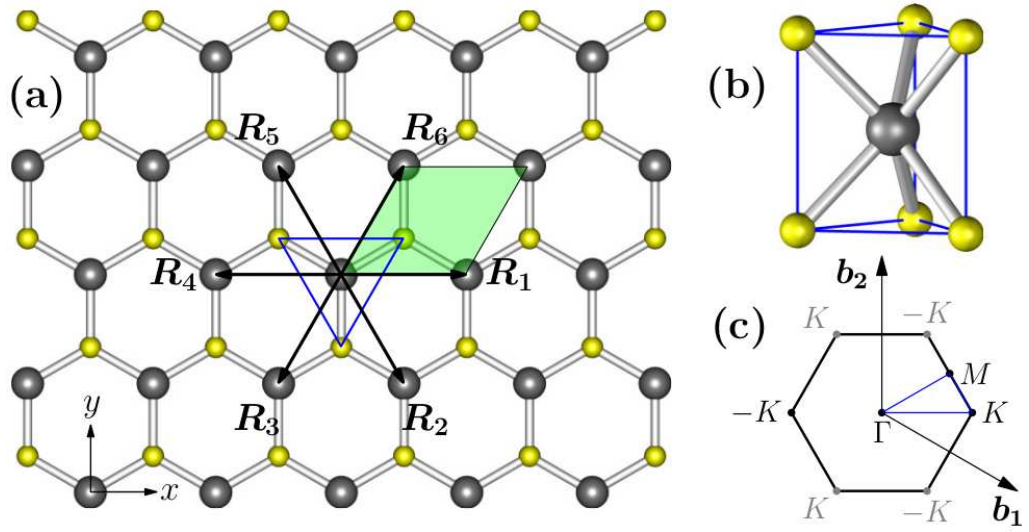


Figure 2: **a** Plan view schematic of a monolayer TMDC with chalcogen atoms shown as small yellow spheres and transition metal atoms shown as large grey spheres. The 2D unit cell is highlighted green and has lattice constant a . $\mathbf{R}_1$ to $\mathbf{R}_6$ shows the nearest neighbour transition metal atoms to the centre transition metal atom. **b** Oblique view of a metal atom with trigonal prismatic coordination to six nearest neighbour chalcogen atoms. The location of such a site is denoted by the blue triangle in A. **c** The Brillouin zone with basis vectors $\mathbf{b}_1$ and $\mathbf{b}_2$. The two valleys K and K^- are shown.

known as valley degree of freedom or valley pseudospin). In addition to this, semiconducting TMDCs show strong spin-orbit coupling which arises from the valence electron states originating from d-orbitals. This results in spin up and spin down carriers having different energies as spin degeneracy is removed from the conduction and valence bands.

These phenomena manifest themselves in the form of optical selection rules in semiconducting TMDC monolayers (though not present in bilayers or thicker). As with normal semiconductors, charge carriers can be promoted from the valence band to the conduction band. However, as previously discussed, the spin states in each band are different for the K and K^- valleys and

can be selectively excited using left- and right-handed circularly polarised light. This selectivity arises from angular momentum conservation and time reversal symmetry constraints.

Promotion of carriers in monolayer semiconducting TMDCs cause electron hole pairs to form strongly bound exciton states due to the enhanced Coulomb interaction between the carriers that results from 2D confinement. In 3D materials exciton states are easily disrupted by phonons and require cooling to low temperatures before they are observed.[ref] However in monolayer semiconducting TMDCs they are detectible at room temperature due to their strong binding energy.

These unique properties of semiconducting TMDCs position them favourably for future optoelectric applications due to their favourable band gap.

Cover metallic and superC TMDCs.

2.1.7 van der Waals' heterostructures

2.1.8 Potential applications for graphene and van der Waals' heterostructures

2.2 (S)TEM imaging of 2D materials

2.2.1 TEM imaging

2.2.2 STEM imaging

2.2.3 Analytical spectroscopic methods: EDXS and EELS

2.2.4 Advanced signal processing techniques for spectrum image datasets

2.2.5 Electron beam induced effects: Damage, heating and contamination

2.2.6 Cross sectional (S)TEM imaging of 2D materials

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